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Ingolstadt

Bachelor Thesis

Faculty of computer science

**Development of a Frontend Application for Real-Time and
24-Hour Monitoring and Recording of Electrocardiographic Data
with Communication via Bluetooth Low Energy**

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AFFIDAVIT

I hereby declare that I have written this thesis independently, have not used any sources or aids other than those stated, have not submitted it elsewhere for examination purposes and have expressly identified all quotations from the sources used, either verbatim or in spirit. I have marked verbatim and indirect quotations as such.

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ABSTRACT

The human heart is the central organ of the circulatory system and plays a vital role in supplying the body with oxygen and nutrients. Rhythmic contractions pump blood through the body, laying the foundation for all vital functions. An essential part of medical diagnostics combined with heart function and health is the electrocardiogram (ECG). The ECG measures the electrical activity of the heart and provides valuable information about the heart rhythm as well as possible irregularities or diseases. As technology has advanced, the way in which ECG data can be collected and monitored has also evolved significantly. Mobile health monitoring in particular now makes it possible to record ECG data not only at certain points of time, but also in real time and over longer periods of time. Wireless transmission via Bluetooth Low Energy (BLE) enables reliable and energy-efficient transfer of data to mobile devices. This opens up new possibilities for applications offering real-time and long term monitoring enabling users access to relevant information and graphical processing and alarm functions. This thesis explains the conception and implementation of an Android frontend application, which realises continuous and mobile ECG monitoring. The main programming language used for this project, is kotlin with Jetpack Compose as the Framework. Moreover this dissertation is particularly focused on implementing the communication and data transmission via BLE as well as realising an 24-hour-mode, which might be used for 24-hour data recognition. It also provides an broad insight into the given requirements, designprincipals, and technical challenges which accured during the implementation. Furthermore this theseis offers a glimpse into the potential for modern healthsuveillance.

CONFIDENTIALITYCLAUSE

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GLOSSARY

Eric Horacsek Absolute operation.

ACRONYMS

D Dimensions.

RGB Red Green Blue channels.

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1 INTRODUCTION

This chapter will give a small introduction on what the thesis and the implemented application is about. This covers an explanation for the application's motivation and the environment the application will be used in.

1.1 Motivation

The aim of this work is to create a new version of an Android App once developed by fellow students from the faculty of Electrical and Information Technology. It's core functionality is to receive and display ECG data in real time, via BLE. With the completion of this work, the new version will be able to receive and display ECG data in real time, via BLE. In addition it will be capable of an 24-hour mode which records ECG data for up to 24 hours, so the collected data can be analyzed later.

1.2 Scope and Framework

This paper only deals with the app development process of the new version. Therefore, any implementation of the previous work done by the students is not going to be discussed in detail. The hardware implementation is also not an element of this work . Generally speaking, this paper covers all topics that are essential for the development of the new app version. Specifically, technology stack of choice, IDE choice desicion, communication with hardware via Bluetooth low energy, structure of the user interface, real-time data monitoring, implementing the 24 hour mode. In addition, the outlook explains procedures for the implementation of future features that could not be realized within the limits of this paper.

1.3 Procedure

The second chapter, *Technical Fundamentals*, explores the key technologies chosen for this project, including the selected programming language, the integrated development environment (IDE), and their respective versions. Additionally, it provides a brief introduction to the Android architecture, with a particular focus on the Java Framework API. This chapter offers valuable insights into the core functionality of the operating system, laying the groundwork for a better understanding of the implementation details presented later. Chapter three delves into the functional requirements, systematically breaking them down into distinct sub-requirements. These requirements are revisited in chapter four, where they are translated into actual code, accompanied by relevant code snippets to illustrate their implementation.

2 TECHNICAL FUNDAMENTALS

2.1 Basics of Android architecture

To program a platform-specific application, it can be important to understand the basic architecture of the platform, especially when dealing with tasks regarding memory or Bluetooth. This section will give a brief insight into the structure of the Android construction. Tasks discussed later in this thesis may reference this chapter to convey a better understanding.

The core of the Android Platform is a Linux based kernel. On top of Android's Kernel are four more stack layers, such as the Hardware Abstraction Layer, Native Libraries/Android Runtime, Java API Framework and the uppermost layer, the System App layer. (also see: Figure 1). Going into detail for each stack layer is not necessary for this work, consequently only essential layers like the Java API Framework will be considered in more depth. A short introduction into each layer is given by table 1 and figure 1 .

Table 1: Layers explanation

Layer	Functionality
System App's (Application Layer)	This layer is a collection of preinstalled applications. Apps contained in this collection, have the same status at system level as third-party apps(system settings is the only exception). Giving the user freedom for design.
Java API Framework (Framework Layer)	Enables acces to all features of the Android OS, such as resource manager and view system.
Native Libraries	System components such as ART and HAL are written in C/C++, those functionalities can be used via API's
Android Runtime (ART)	Every running application is an instance of ART. ART operates on the same layer as Native Libraries
Hardware Abstraction Layer (HAL)	Provides interfaces to the Java API Framework, hence addressing hardware components like Camera or Bluetooth module is possible.
Linux Kernel	Foundation of the OS. Administrates system resources, and communicates with hardware.

A layered archtitecture like the Android OS, ensures modularity, maintainability and security. Each layer is designed with a specific responsibility, and communicates primarily with adjacent layers. Applications interact with the Framework layer via the Android SDK (bridge between Application Layer and Framework Layer), which provides access to core system services. The Framework, in turn, relies on the Native Libraries and Runtime layers for functionalities like graphics rendering, media playback, and app execution.

Hardware communication is abstracted through the Hardware Abstraction Layer (HAL), which interfaces with the Linux Kernel. This structured separation of concerns enhances the system's flexibility, allowing independent evolution and optimization of each layer.

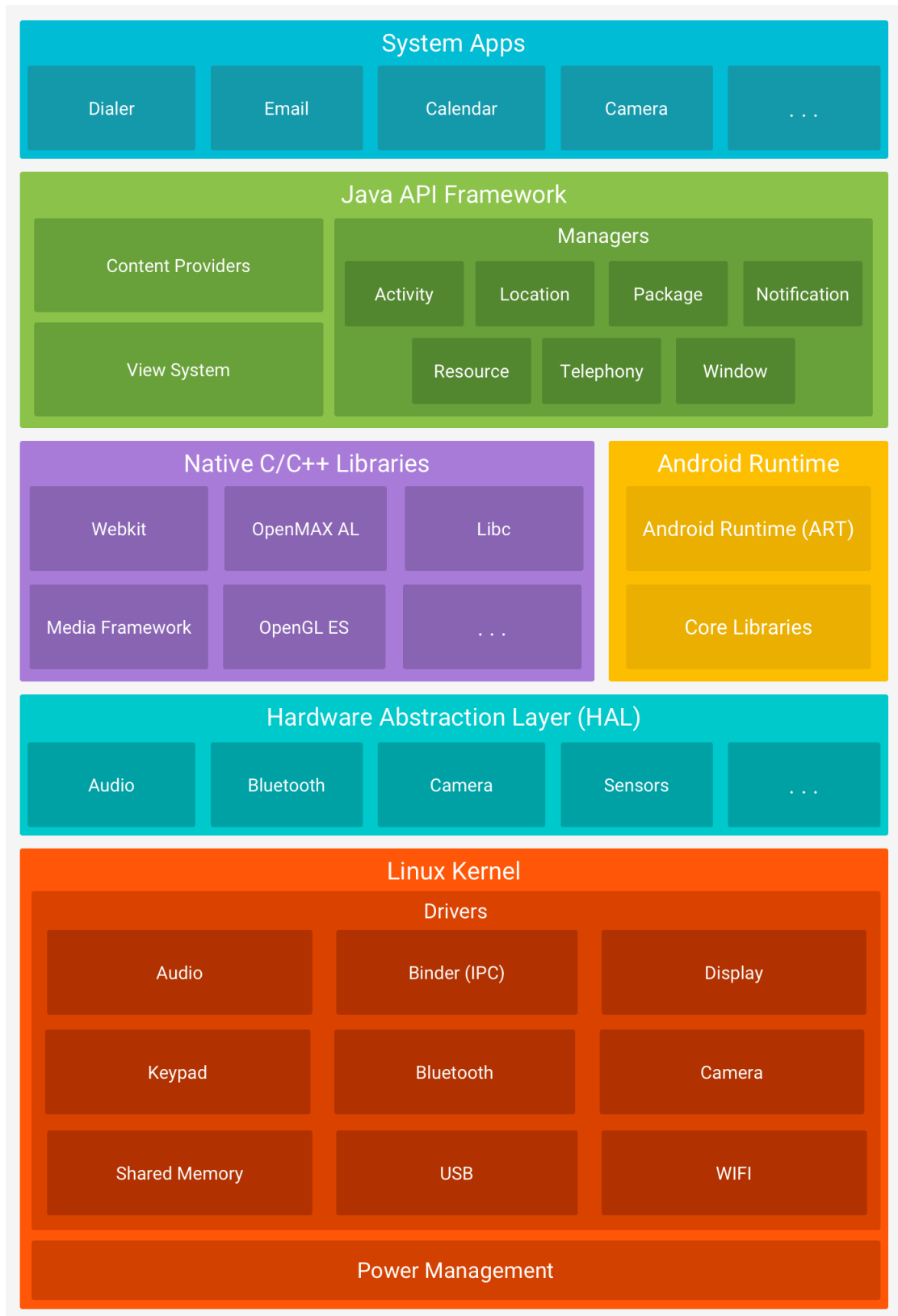


Figure 1: Android Platform architecture

2.1.1 Java Framework API (Framework Layer)

Description

As already mentioned in 2.1, the Android Java Framework API is a collection of libraries and tools provided by Android to simplify the development of mobile applications. Built on top of Java, it enables developers to create apps with pre-built components, utilities, and services that interact with the Android operating system. The framework abstracts the complexity of low-level operations and provides high-level APIs for features such as UI design, data storage, networking, and hardware access.

Project example

To provide an impression, Figure 2 shows a code excerpt in which the Java API was implicitly used by androidx. To keep this chapter manageable, it can be simplified said that androidx is a superset of the Java API Framework.

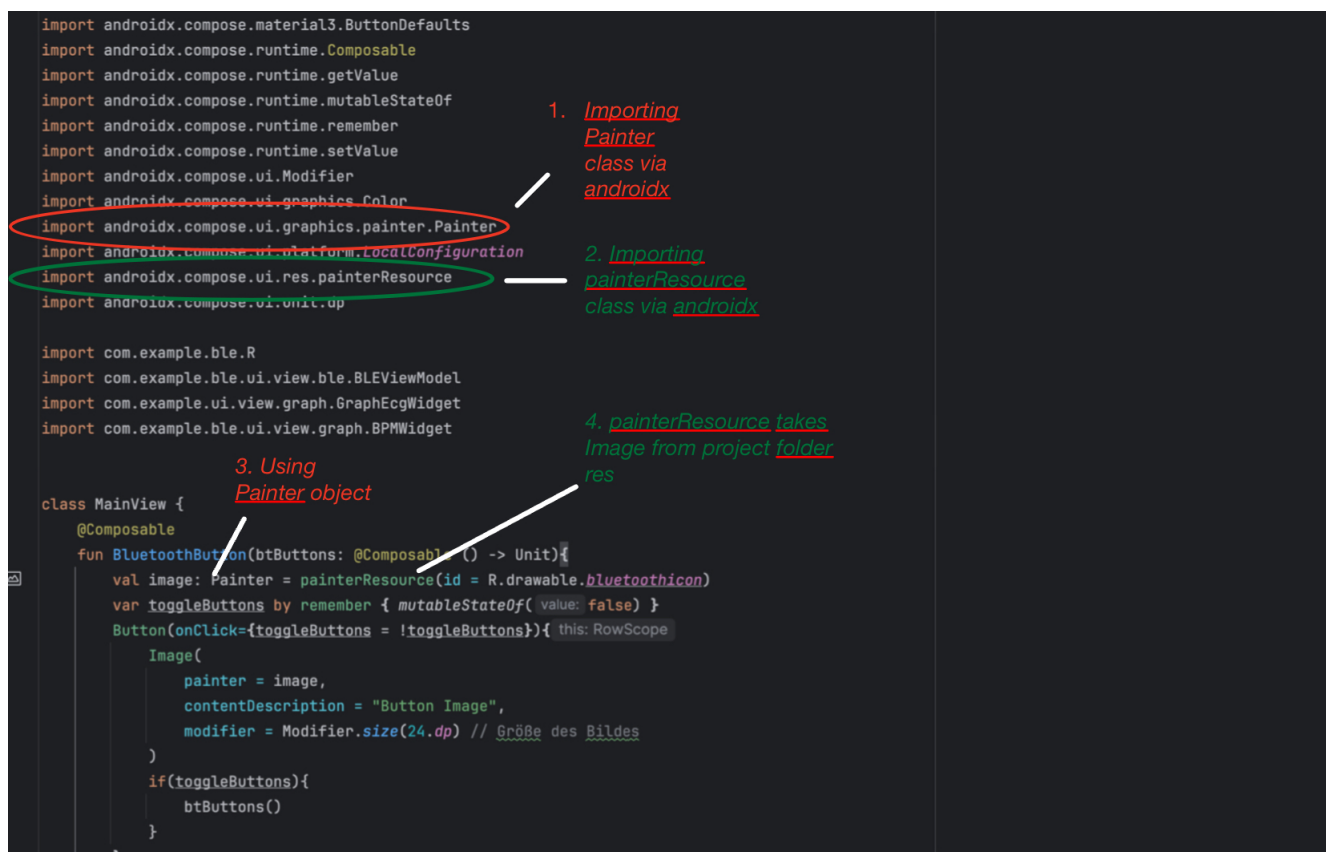


Figure 2: Android Java API example with androidx

Figure 2 provides a glimpse of the capabilities of the androidx library. Showing how to draw an image with the Painter class and utilizing the resource manager mentioned in table 1.

To begin its elementary to import the Painter class aswell as the painterResource class. As shown in 1. and 2. in Figure 2. Then at 3. a Painter object is created, at 4. the Painter object is defined by the painterResource function. The painterResource function gets an id as an argument, to query the file from the res/drawable folder. The res folder contains all additional app resources (files, images, etc.). Finally we can draw the Image by passing the object from 3. to an Image function.

3 FUNCTIONAL REQUIREMENTS

Requirement engineering is essential at the beginning of every project. Formulating and documenting precisely defined requirements is a cornerstone in the course of realization. In this way, mundane mistakes can be avoided and the vision of the final product is clear. Thus this chapter will explain all functional requirements of the finished product.

3.1 ECG-Data

The application receives integer values from the microcontroller, which represent the heartbeat of the person connected to the hardware. These values must be mapped onto the screen to provide a clear and accurate visualization of the heart's activity. A key challenge in this process is ensuring that the visualization adapts to different screen sizes. Since the application may run on devices with varying display dimensions and resolutions, a static mapping approach is insufficient. Instead, a dynamic and scalable algorithm is required to transform the received data into a graphical representation that maintains consistent proportions and readability across all screen sizes. To achieve this, the application must implement a flexible coordinate transformation. The incoming data points need to be scaled appropriately based on the available screen width and height while maintaining the correct aspect ratio of the graph. Additionally, the algorithm should support real-time updates to provide an accurate and responsive user experience. The transformation of raw data points must be performed quickly, otherwise, there is a risk of data loss. Therefore many parameters are need to be considered when transforming the data.

Parameters

- Screen dimensions (height, width)
- Current min and max value in buffer
- Buffer capacity

- Δv , where $\Delta v = \max - \min$ (range of values in current Buffer)
- coefficient = $\frac{\text{height} - 2 \times \text{shiftH}}{\Delta v}$ (maps raw values to screen height, with margin of shiftH pixels at the top and bottom)
- step = $\frac{\text{width}}{\text{bufferCap.}}$ (Determines horizontal space for each data point)

After calculating these parameters, each of the current values in the buffer will be multiplied by the Scaling coefficient ($\frac{\text{height} - 2 \times \text{shiftH}}{\Delta v}$).

Resulting in the following Complexity Analysis:

- Data Conversion: $O(n)$ (for n data points)
- Scaling & Mapping: $O(n)$ for n data points)
- Overall Complexity: $O(n)$ linear (for n data points)

The resulting linear Complexity of $O(n)$ makes the algorithm efficient for real-time processing.

3.2 Drawing ECG-Data in realtime

Processing real-time electrocardiographic (ECG) data presents multiple challenges that must be addressed to ensure accurate and efficient visualization. It is helpful to split this rather complex functionality into multiple concerns:

- The sample rate of the microcontroller (1)
- Consume and process incoming data simultaneously (2)
- Datahandling and Drawing in parallel (3)

These issues must be viewed and solved isolated, and subsequently combine each solved problem with the other. It is important to mention that some issues may create restrictions for other issues. Therefore, it is essential to implement one task at a time and connect them all afterward.

(1) Handle The Sample rate of the microcontroller

The microcontroller sends 10 samples every 8 milliseconds ($1000\text{ms} = 1\text{s}$).

$$\frac{1000}{8} = \frac{125}{1} \hat{=} 125\text{Hz}$$

Hence we have to process a sample rate of: $f(t) = 125 \times t$, t in seconds. If handling this datastream is not possible, data will be lost.

(2) Consume and process incoming data at the same time

To consume and process data at the same time, splitting up processing and consuming into two indepent tasks by assigning each task to its own thread, is necessary. Making it possible to operate in paralell.

Since both threads have to access the same ressource at the same time, we have to make use of mutual exclusion.

(3) Datahandling and drawing in parallel

Thus the drawing entity has to countiously read data, and the datahandler entity receives a continous data stream. The suitable data structure must be able to constantly record new data and retain a certain amount of old data. So that a screen-width drawing can be displayed continuously. In addition, it is important that the data structure is not to large, which would lead to visible latencies.

This led to the Ringbuffer beeing the datastructure of choice. Ringbuffers are often used when having to process datastreams. The ringbuffer has a fixed size, thus old data gets overwritten. Therefore, there is no need to worry about the data structure consuming too much memory.

3.3 24-Hour recording

The functional requirements of the 24-Hour recording are shown in the list below. All of the listed requirements will be explainend in detail in chapter 4.

- Recording mode
- Filechooser concept, to choose multiple files
- Start a recording, and assigning a name
- Abort a recording
- Display 24-Hour long recorded ECG-Data
- Show the current time intervalls
- Zoom into the displayed data, zooming out of the displayed data

- Plot BPM of the current displayed time intervall

3.4 Connection via Bluetooth Low Energy

The application is required to establish and maintain a reliable Bluetooth Low Energy (BLE) connection with the hardware. A stable connection is essential for continous data-transfer aswell as for the 24-Hour recording Mode. The BLE entity must be capable of the following items:

- Continous datastream
- Communicate with the application
- Share current the current mode (realtime, or recording)
- Scan for near devices
- split & reassamble to large datasets

4 REALIZATION

The first part of this chapter provides an overview of the information flow between each entity, and the tasks of the given entitys. Aswell as an close look into the implementation of the functional requirements, by analysing code snippets, aswell as explaining the usage of the final product. While the second part will explain every necessity to assemble all parts together for a working product.

4.1 Selected Technologies

"The nature of the application and the required functionality of a software system should influence the choice of methods and tools used during development". [Clarke \[1995\]](#) This ensures that the development process is both efficient and tailored to meet the unique demands of the system. Moreover, aligning tools and methods with the application's requirements can help streamline workflows and reduce potential complexities during implementation. Consequently choosing the right programming language, aswell as IDE is essential to enable efficient development.

4.1.1 Programming Language and Framework

When searching for a suitable programming language to program Android apps, names like Java/Kotlin and Dart often appear. Both languages offer great frameworks, such as Flutter for Dart or Jetpack Compose for Kotlin. One as well as the other are highly effective because they eliminate the limitations of XML-based UI development, enabling a more streamlined, declarative approach that combines UI and logic in a single language. Thus reduces boilerplate code, and enhances performance as well as developer productivity. The main difference between both frameworks lies in resource management. For example Flutter is superior in CPU usage, but Jetpack Compose consumes less memory. [Kusuma et al. \[2023\]](#) "But since both have their respective advantages and disadvantages, it is impossible to say that one is superior to the other. The choice between Jetpack Compose and Flutter depends on the needs and preferences of the developer. If developers are more familiar with the Kotlin language and want to use new technologies, then Jetpack Compose might be a better choice." [Kusuma et al. \[2023\]](#) With prior practical experience in Java/Kotlin, Jetpack Compose was chosen as the framework for this work.

4.1.2 Development Environment

Integrated development environments play a fundamental role in today's application development, providing developers with tools mandatory to debug, refactor and write code. Making programming more efficient and delightful for programmers. When developing Android Apps with state-of-the-art frameworks like Jetpack Compose, Android Studio stands out as the most suitable option as both Jetpack Compose and Android Studio are created by Google. Android Studio is the official IDE for Android development. With IntelliJ as a base it inherits powerful code editor and developer tools, combined with Android-specific features. Such as Live Edit (updates running app when code is changed) as well as Interactive Preview which allows to test functionality between so-called Compose components (Compose will be explained in the later course of this work)¹, et cetera.

The version of Android Studio used for this thesis is **Ladybug | 2024.2.1**. Which is the latest version as this was written (13.12.2024).² Downloading the right version can be crucial for problem-free debugging. As some versions lead to bugs, causing the screen to freeze.³

¹<https://developer.android.com/studio/intro>

²<https://developer.android.com/studio/releases>

³<https://stackoverflow.com/questions/78125635/processing-classes-for-emulated-method-breakpoints-this-popup-comes-and-loadin>

4.2 Definiton of Entitys

As shown in table 2 each entity has its scope of dutys, making the final code more modular and consequently maintainable.

Table 2: Entiys area of responsibility

Layer	Functionality
BLEViewModel	- Communication with the microcontroller - Receives datastream - Sends instructions to microcontroller
GraphEcgWidget	Draws the ECG data in realtime, and displays Beats per minute
Storewrapper	Converts data from microcontroller into drawble data
BPM	Calculates the current Beats Per Minute
VitalMonitorWidet	Plots up to 24 Hour long recoreded ECG data and the BPM of the plotted Intervall. Provides zooming functionality.
StorageHandler	Allows usage of the OS filechooser, to select files.
MainView	Collection of visble elements (Widgets), to provide reusability
MainActivity	Visible to user, contains the complete User Interface. Takes and processes user inputs.

4.3 Inter-Entity Communication Overview

The following diagram shows the Information flow between all entities. The unidirectional arrow means that, information is flowing on one side. An information exchange between two entities is represented by a bidirectional arrow.

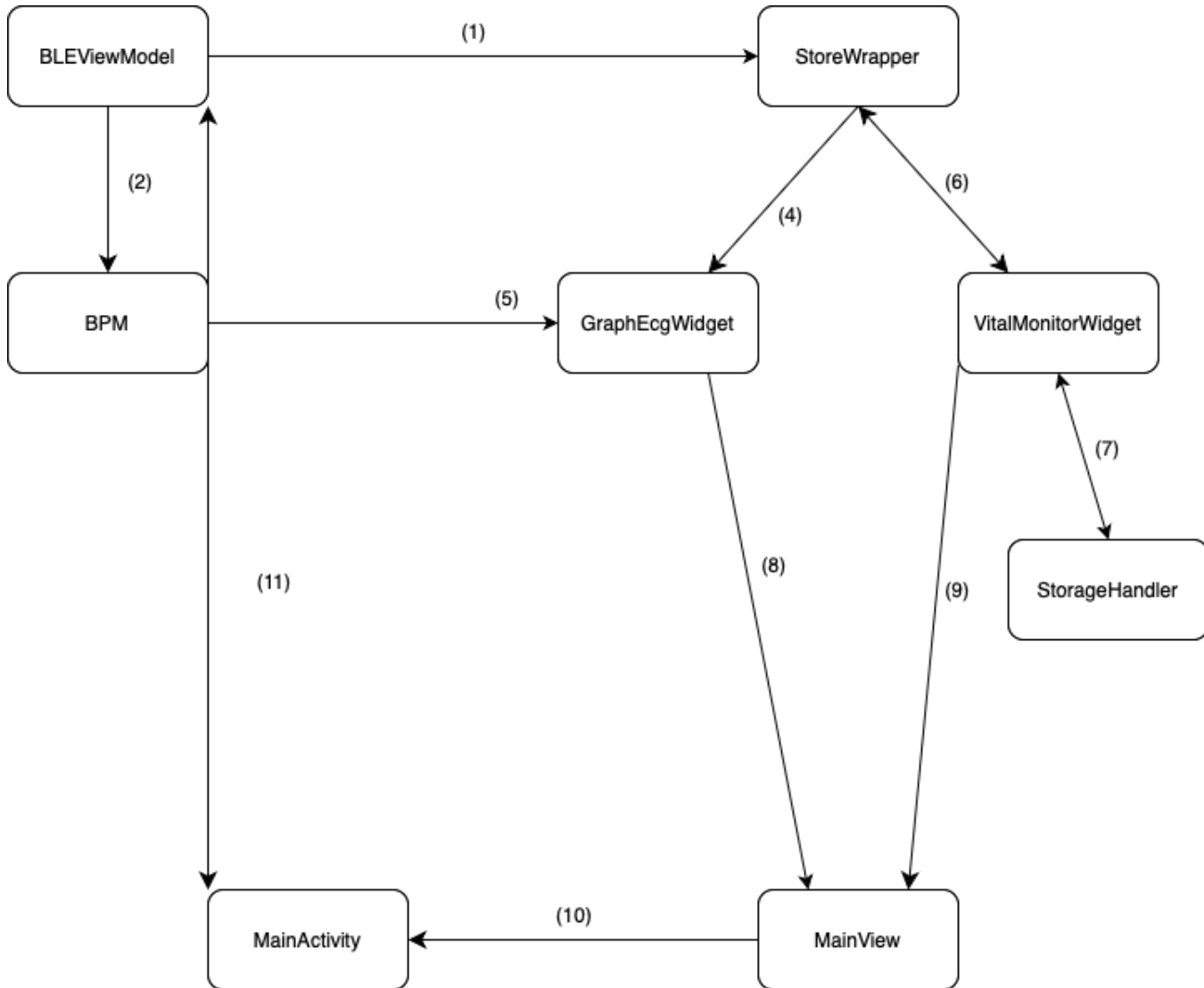


Figure 3: Informationflow Diagramm

As already stated in 4.2 the BLEViewModel entity, serves as an interface between controller and app. Therefore every entity that needs to evaluate recorded data, is dependent on this Instance. Two of the entities listed in 4.2 are affected. To be more precise, it is the StoreWrapper that makes the data drawable (1) and forwards the data to the GraphECGWidget (4). The BPM instance that calculates the beat frequency (2), (which

is then displayed by the GraphECGWidget (5)). When comparing Informationflow (4) and (6) both seem very similar, because both widgets obtain drawable data. However the Informationflow of (6) is bidirectional. This is necessary because the recorded data needs to be read, by the StorageHandler from a local file (7), before being sent to the Storewrapper (6). While in (4) the Storewrapper receives data directly from the ECG. After the data has been mapped by the Storewrapper, it is sent back to the VitalMonitorWidget (6). MainView integrates both the GraphECGWidget (8) and the VitalMonitorWidget (9) as components and manages their visibility. However, the range of tasks of the BLE-ViewModel is not limited to exchanging data, but also extends to scanning and connecting with other devices in the vicinity, as well as integrating and managing a 24-hour ECG recording. These functionalities are utilized by the MainView and the MainActivity. As already mentioned in table 2, MainView serves solely to construct components of functionalities that are later utilized by MainActivity, such as buttons enabling users to start a 24-hour ECG recording (10). The MainActivity entity acts as a bridge between the microcontroller and the user, allowing device scanning, initiating recordings, establishing connections, and displaying ECG information (11).

Information Flow and Responsibilities of BLEViewModel

- **BLEViewModel as an Interface**
 - Acts as an interface between the controller and the app (4.2).
 - Any entity that evaluates recorded data depends on this instance.
- **Entities Dependent on BLEViewModel**
 - **StoreWrapper**
 - * Makes the data drawable (1).
 - * Forwards the data to the **GraphECGWidget** (4).
 - **BPM Instance**
 - * Calculates the beat frequency (2).
 - * The result is displayed by the **GraphECGWidget** (5).
- **Comparison of Information Flows (4) and (6)**
 - Both provide drawable data to widgets.
 - **Difference:**
 - * **(4) One-way flow** → StoreWrapper receives data directly from the ECG.

-
- * **(6) Bidirectional flow** → Required for reading recorded data from a local file (7) using the **StorageHandler** before sending it to the **StoreWrapper**.
 - Once mapped by the **StoreWrapper**, the data is sent back to the **VitalMonitorWidget** (6).
 - **MainView Responsibilities**
 - Integrates **GraphECGWidget** (8) and **VitalMonitorWidget** (9).
 - Manages the visibility of these components.
 - **Extended BLEViewModel Responsibilities**
 - Beyond data exchange, **BLEViewModel** also:
 - * Scans and connects to other nearby devices.
 - * Integrates and manages 24-hour ECG recordings.
 - These functionalities are used by **MainView** and **MainActivity**.
 - **MainView vs. MainActivity**
 - **MainView**
 - * Constructs components and functionalities for **MainActivity**.
 - * Provides UI elements, such as buttons for starting a 24-hour ECG recording (10).
 - **MainActivity**
 - * Acts as a bridge between the **microcontroller** and the **user**.
 - * Handles device scanning, recording initiation, connection establishment, and ECG data display (11).

4.4 BLEViewModel

The following subsections explore the logical implementation and functionality of individual entities, using code snippets and application screenshots for illustration. Each subsection is divided into a section covering the implementation and another focusing on the application's usage, with both sections referring to corresponding screenshots. Framework specific code fragments will be explaining in the course of relevant section.

Before presenting the implementation of the BLE component, it is essential to establish a fundamental theoretical understanding of BLE. This section offers a comprehensive

explanation of the Bluetooth Low Energy, shedding light on its core principles, while exploring the architecture of the BLE stack.

4.4.1 BLE Stack

The BLE protocol stack consists of three main layers: the Controller, the Host, and the Application. The Controller encompasses the Physical Layer and the Link Layer, while the Host manages higher-level functionalities such as the Logical Link Control and Adaptation Protocol (L2CAP), the Attribute Protocol (ATT), the Generic Attribute Profile (GATT), the Security Manager (SM), and the Generic Access Profile (GAP). Communication between the Host and the Controller follows the standardized Host Controller Interface (HCI). At the top of the stack, the Application layer enables program execution.

Dian et al. [2018]

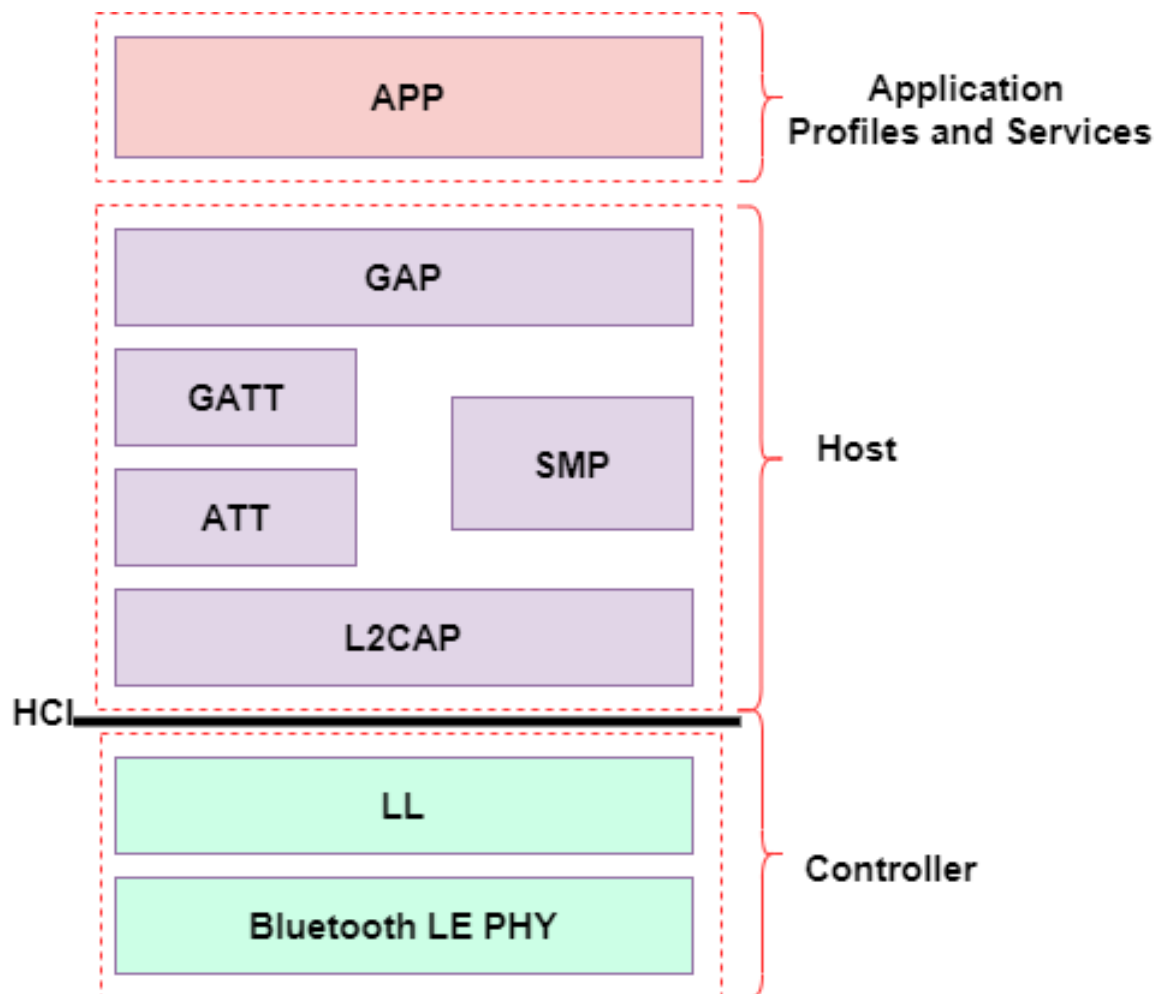


Figure 4: BLE protocol stack (<https://de.mathworks.com/help/bluetooth/gs/bluetooth-protocol-stack.html>)

Application Layer

The application layer serves as the direct user interface, defining profiles that enable interoperability between various applications.

Host Layer

Includes the host-sided HCI as well as L2CAP, ATT, GATT, SMP and GAP.

Generic Access Profile (GAP)

Manages the connection setup as well as security, interfaces directly with profiles and services of the Application Layer.

Security Manager Protocol (SMP)

Provides multiple security algorithms for encrypting and decrypting data packets, making the connection secure.

Generic Attribute Profile (GATT)

Defines services and characteristics provided by the GATT Server. The GATT Server in this case is the microcontroller, which holds the definitions of services and characteristics. These can be requested by the GATT Client (in this case the application). Services organize data into logical entities and contain specific data units known as characteristics. Each service can include one or more characteristics and is uniquely identified by a numeric UUID. This UUID can be either 16-bit for officially adopted BLE services or 128-bit for custom services. Whereas the Characteristic is the lowest level concept in GATT transactions, which defines data. Services as well as Characteristics are main interaction points to work with when working with the BLE periphery, thus it is important to understand these terminologies.

Attribute Protocol (ATT)

The ATT enables the exchange of attributes (characteristics). Attributes are composed of four components, the type (128-bit UUID), an handle (16-bit identifier) to reference attributes, rights of the attribute (reading, writing), and the value of the attribute. The following figure gives a good overview of the structure of the attributes.

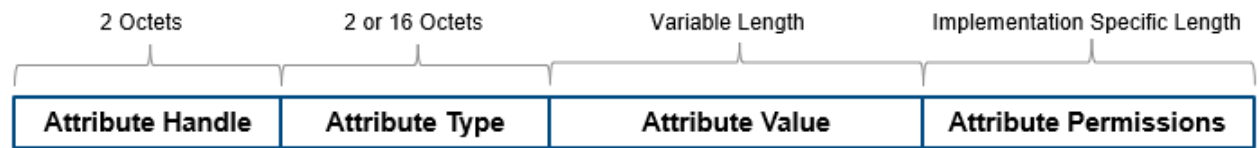


Figure 5: Attribute Structure (<https://de.mathworks.com/help/bluetooth/gs/bluetooth-protocol-stack.html>)

Logical Link Control and Adaption Layer Protocol (L2CAP)

Works as an interface between the HCI and the upper modules of the Host layer, by multiplexing data packets to the corresponding protocols. Also fragments and reassembles large packets.

Host-Side Host-Controller-Interface (HCI)

Generally said the HCI is the interface between the host and the controller. The HCI establishes commands and events for packet data transmission and reception. When sending data, it processes raw information into packets, enabling communication between the host and the controller.

Controller-Side Host-Controller-Interface (HCI)

As mentioned in the last paragraph, the HCI is the interface between the host and the controller. However, analogous to the host side, the data received from the controller is extracted and forwarded to the host.

Controller Layer

Includes the Bluetooth LE PHY, LL, and controller-side HCI

Link Layer (LL)

The LL functions similarly to the medium access control (MAC) layer in the OSI model. In Bluetooth, it directly interacts with the Bluetooth LE PHY and controls the radio's link state, determining whether a device operates as a Central, Peripheral, Advertiser, or Scanner.

Bluetooth LE PHY

The physical layer (BLE PHY) is a 1 Mbps GFSK-based radio with adaptive frequency hopping, operating in the unlicensed 2.4 GHz ISM (industrial, scientific, and medical) band.

4.4.2 Implementation

Dependency versions and Library

Before delving into the implementation, it is essential to note that a library provided by Nordic Semiconductor was employed in the development process.⁴ The utilized dependency version are shown in the figure below:

```
implementation("no.nordicsemi.android.kotlin.ble.scanner:1.1.0")
implementation("no.nordicsemi.android.kotlin.ble.client:1.1.0")
implementation("no.nordicsemi.android.kotlin.ble.advertiser:1.1.0")
implementation("no.nordicsemi.android.kotlin.ble.server:1.1.0")
implementation("no.nordicsemi.android.ble:2.4.0")
implementation("org.jetbrains.kotlinx:kotlinx-coroutines-core:1.6.0")
implementation("io.coil-kt:coil-compose:2.2.2")
implementation("androidx.collection:collection:1.4.3")
```

Figure 6: Library Versions

Scanning devices

The following code snippet demonstrates the implementation of a BLE scanning method used to detect nearby devices. The *startScan* function initializes the scanning process by setting the *scanState* to *SCANNING* and resetting the list of discovered BLE devices. The scanning operation is performed using *BleScanner*, with results processed by *BleScanResultAggregator*. The detected devices are filtered to include only those with the name *"THI-EKG"* and stored in a *MutableState*. The scan runs asynchronously within the *viewModelScope*, and after a delay of 2000 milliseconds, it is terminated by canceling *scanJob* and updating the *scanState* to *DONE*. The snippet also contains two annotations above the function definition, which don't add any functionality, but provide robustness and readability to the code. Nevertheless, a brief explanation follows to avoid confusion and ensure clarification:

@RequiresApi(Build.VERSION_CODES.UPSIDE_DOWN_CAKE)

This annotation ensures that the *startScan* function is only executed on devices running Android API level **34 (Android 14, Upside Down Cake)** or higher. If the function is called on a lower API level, the compiler will issue a warning, and runtime execution may fail.

@SuppressLint("MissingPermission")

This suppresses lint warnings related to missing Bluetooth permissions (such as *BLUETOOTH_SCAN* or *ACCESS_FINE_LOCATION*). Normally, scanning for BLE devices requires specific permissions, and Android Studio would generate a warning if they are not

⁴<https://github.com/NordicSemiconductor/Android-BLE-Library>

explicitly checked. However, this annotation tells the lint tool to ignore such warnings, assuming the necessary permissions are already handled elsewhere in the code. Thus the permissions are checked in the MainActivity entity, this warning can be ignored, and will appear more often.

```
private var scanJob: Job? = null

@RequiresApi(Build.VERSION_CODES.UPSIDE_DOWN_CAKE)
@SuppressLint("MissingPermission")
private fun startScan(context: Context) {
    scanState.value = ScanState.SCANNING
    foundBLEDevices.value = mutableSetOf()

    val aggregator = BleScanResultAggregator()
    scanJob = BleScanner(context).scan()
        .map { aggregator.aggregateDevices(it) }
        .onEach { it: List<ServerDevice>
            foundBLEDevices.value = it.stream()
                .filter { it.name == "THI-EKG" }
                .collect(Collectors.toSet())
        }
        .launchIn(viewModelScope)

    viewModelScope.launch { this: CoroutineScope
        delay( timeMillis: 2000)
        scanJob?.cancel()
        scanState.value = ScanState.DONE
    }
}
```

Figure 7: Scanning functionality code

Refreshing

Before initiating the BLE scan, the function checks for two critical permissions:

- BLUETOOTH_SCAN: This permission is required to scan for nearby BLE devices.
- BLUETOOTH_CONNECT: This permission is necessary to establish a connection with discovered BLE devices.

The function uses `ActivityCompat.checkSelfPermission` to verify whether these permissions have been granted by the user. This step is crucial for ensuring that the application complies with Android's permission model, which mandates explicit user consent for accessing sensitive device features. Conditional Execution: If both permissions (`BLUETOOTH_SCAN` and `BLUETOOTH_CONNECT`) are granted, the function proceeds to call `startScan(context)`. If the permissions are not granted, the function includes a `TODO` comment, indicating that additional logic is required to handle this scenario. Typically, this would involve prompting the user to grant the necessary permissions through a permission request dialog.

```
@RequiresApi(Build.VERSION_CODES.UPSIDE_DOWN_CAKE)
fun refreshBLE(context: Context) {
    if (ActivityCompat.checkSelfPermission(
        context,
        Manifest.permission.BLUETOOTH_SCAN
    ) == PackageManager.PERMISSION_GRANTED &&
        ActivityCompat.checkSelfPermission(
            context,
            Manifest.permission.BLUETOOTH_CONNECT
        ) == PackageManager.PERMISSION_GRANTED
    ) {
        startScan(context)
    } else {
        //TODO: Handle the case where permissions are not granted
    }
}
```

Figure 8: Refresh functionality code

Connect

The provided BLE implementation is responsible for establishing a connection with a BLE-enabled device, receiving real-time data from specific BLE characteristics, and processing the incoming data to extract and analyze electrocardiographic (ECG) information. The functionality can be divided into the following key components:

1. BLE Connection Setup (Line 143-149)

The function `connect` initializes the connection with a given BLE device. Using `ClientBLEGatt.connect`, a GATT connection is established, and the discovered services and characteristics are retrieved. The relevant characteristics include:

signalCharacteristic: Used for receiving real-time electrocardiographic data.

appCommunicationCharacteristic: Used for bidirectional communication with the BLE

device. The obtained `appCommunicationCharacteristic` is stored in the `BLEViewModel` for further use.

2. Receiving and Processing Data from the Signal Characteristic

The `signalCharacteristic`, continuously provides data in the form of a `ByteArray`. Each received data packet undergoes the following procedure:

The received byte array is first converted into a list of integers. To enable concurrent processing, the data is then transmitted through a channel (`channel.send(values)`). A coroutine, executed by `receiveData(viewModelScope)`, retrieves the streamed data from the channel and passes it to `StoreWrapper.processDataList(datastream)`. (Since coroutines and channels play a central role in this and other implementations, this chapter is followed by a brief introduction to these Kotlin-specific topics.) The processed data is also added to the `BPM.currList` buffer for frequency analysis. Once the buffer exceeds 7500 samples, the QRS peak detection algorithm (Pan-Tompkins) is applied to identify heartbeats. The detected peak count is logged for debugging, and to maintain a sliding window approach, the oldest 750 samples are discarded.

3. Handling Application-Level BLE Communication

Notifications are also enabled for the `appCommunicationCharacteristic`, which allows bidirectional communication with the BLE device. Each received notification undergoes:

Validation using `ResponseChecker.check`. Packet reconstruction via `PacketReassembler.reassemble`. Logging of the received data for debugging purposes.

4. Additional Commands

The implementation also includes:

Sending an MTU request (`requestMtu(512)`) to maximize packet size for efficient data transmission. Checking if the application is in live mode or in 24-Hour mode by calling `fetchCurrentMode()`. Also fetching all recorded ECG's from the microcontroller `ListEKGS()`.

Channels and Coroutines

The *channel* serves as a concurrent data pipeline between the BLE data stream and the processing logic. It ensures that the incoming data from the BLE device is processed in a structured, thread-safe manner without blocking the main execution thread. By leveraging coroutines, the channel allows efficient real-time data handling while preventing performance bottlenecks.

A *coroutine* in Kotlin is a lightweight, non-blocking way to handle asynchronous programming. It allows you to write code that can be paused and resumed without blocking the main thread, making it ideal for handling tasks like network requests, BLE communication, or large computations without freezing the UI.

```

140 @SuppressLint("MissingPermission")
141 fun connect(context: Context, device: ServerDevice) {
142     this.viewModelScope.launch { this: CoroutineScope
143     ->         val connection = ClientBleGatt.connect(context, device, this@BLEViewModel.viewModelScope )
144         this@BLEViewModel.connection.value = connection
145     ->         val services = connection.discoverServices()
146         val bleService = services.findService(UUID.fromString( name: "e28e00e9-b79a-479a-a6f1-21e8f73236e1"))
147         val signalCharacteristic = bleService!!.findCharacteristic(UUID.fromString( name: "23b85b26-e7cb-421c-9731-1b4ca2d2a849"))
148         val appCommunicationCharacteristic = bleService!!.findCharacteristic(UUID.fromString( name: "563a2846-1dc7-11ef-9262-0242ac120002"))
149         this@BLEViewModel.appCommunicationCharacteristic = appCommunicationCharacteristic!!
150
151     ->         signalCharacteristic!!.getNotifications().onEach { it: DataByteArray
152             val values = byteArrayToIntArray(it.value).toList()
153             startTime = System.currentTimeMillis()
154     ->             channel.send(values)
155             receiveData(viewModelScope){dataList ->
156                 StoreWrapper.processDataList(dataList)
157                 BPM.currList.value += dataList
158                 if(BPM.currList.value.size >= 7500){
159                     var qrsPeaks = BPM.panTompkins(BPM.currList.value).size
160                     println("PEAK COUNTER: ${qrsPeaks}")
161                     println("liste: ${BPM.currList.value}")
162                     BPM.currList.value.forEach { it: Int
163                         println(it)
164                     }
165                     println("ENDOF PRINTING")
166                     BPM.currList.value = BPM.currList.value.subList(750, 7500-1)
167                     BPM.countPeak = qrsPeaks
168                     qrsPeaks = 0
169                 }
170             }
171             paketCounter++
172         }.launchIn(viewModelScope)
173     ->         appCommunicationCharacteristic!!.getNotifications().onEach { it: DataByteArray
174             ResponseChecker.check(it.value)
175             PacketReassembler.reassemble(it.value)
176             println(it)
177         }.launchIn(viewModelScope)
178     ->         connection.requestMtu( mtu: 512)
179         fetchCurrentMode()
180         listEKGS()
181     }
182 }
183

```

Figure 9: Connect functionality code

4.5 Storewrapper

The StoreWrapper class serves as a central component for processing and managing electrocardiographic (ECG) data in real-time. It is responsible for converting raw data received from the BLE signal characteristic into a format suitable for visualization. This transformation ensures that the incoming stream of ECG values can be efficiently rendered within the application's graphical interface. A key feature of the StoreWrapper class is its mapping functionality, which translates raw numerical values into drawable data points. This conversion is essential for accurately representing the ECG signal on-screen. Additionally, the class incorporates a ringbuffer, which provides a mechanism for storing and handling continuous data streams. The ringbuffer ensures that only the most recent samples are retained, enabling a sliding window approach for real-time signal visualization and processing.

4.5.1 Implementation

Preparing ECG-Data

The provided code snippet is responsible for transforming raw electrocardiographic (EKG) data into a format suitable for visualization across different screen sizes. This transformation ensures that the signal is correctly scaled and displayed within the graphical interface, regardless of resolution or aspect ratio. The function achieves this with a time complexity of $O(n)$, making it efficient for real-time applications.

The function begins by initializing the shiftH parameter and preparing a mutable list (data) for storing processed values. If an external data source is provided, it is converted from a generic format into a list of Double values. Otherwise, an empty list is created.

Screen Size and Scaling Factors

The width and height of the target display area are extracted from the size parameter. Additionally, the function determines the minimum (minV) and maximum (maxV) values within the buffer to establish the dynamic range of the data. If all values are equal, an artificial spread is introduced to prevent division by zero.

Normalization and Scaling

The difference between maxV and minV (dv) is calculated to determine the range of data values. A step size (step) is then computed based on the available screen width and the buffer capacity, ensuring that data points are evenly distributed across the display. The

coeff value is derived to scale the signal to fit within the available height while accounting for a vertical shift (shiftH).

Data Mapping for Graph Rendering

dataSeriesOverlay(buffer_) A temporary list (dataTemp) holds the normalized data, which is then mapped to a new list (data). The mapping follows the formula:

$$data[i] = (maxV - dataTemp[i]) \times coeff + shiftH$$

This equation ensures that the data is flipped and scaled appropriately for graphical representation. The function iterates over the dataset in a single pass (O(n) complexity), making it highly efficient for real-time applications. This guarantees smooth and responsive rendering, even with large data sets.

```

1      fun prepareData(size: Size, shiftH: Double, dataFromFile:
2          MutableList<Any>? = null): List<Double> {
3          this.SHIFTH = shiftH
4          var data: MutableList<Double>? = null
5          if( dataFromFile != null ){
6              data = dataFromFile.mapNotNull {
7                  (it as? String)?.toDouble()
8              }.toMutableList()
9          }else{
10             data = ArrayList()
11         }
12         val width = size.width.toDouble()
13         val height = size.height.toDouble()
14         var minV = min //lowest value in buffer
15         var maxV = max //highest value in buffer
16         if (minV == maxV) {
17             minV = minV / 2
18             maxV = maxV + minV / 2
19         }
20         val dv = maxV - minV
21         step = width / buffer_.capacity()
22         val coeff = (height - 2 * shiftH) / dv
23         //passing the wrong buffer to dataSeriesOverlay (not updated)
24         val dataTemp =
25         if (_mode == GraphMode.overlay) utils_.dataSeriesOverlay(
26             buffer_) else utils_.dataSeriesNormal(
27             this
28         )
29         data = ArrayList(Collections.nCopies(dataTemp.size, 0.0))
30         //return data

```

```

29         if (data.isEmpty() || dataTemp.isEmpty()) {
30             return data
31         }
32         for (i in dataTemp.indices) {
33             data[i] = (maxV - dataTemp[i]!!) * coeff + shiftH
34         }
35         return data
36     }

```

Code 1: Einfaches Beispiel eines Sourcecode-Generators, abgeleitet von ?? und eigenen Ergänzungen

```

167 fun prepareData(size: Size, shiftH: Double, dataFromFile: MutableList<Any>? = null): List<Double> {
168     this.SHIFT_H = shiftH
169     var data: MutableList<Double>? = null
170     if (dataFromFile != null) {
171         data = dataFromFile.mapNotNull { it: Any
172             (it as? String)?.toDouble()
173         }.toMutableList()
174     } else {
175         data = ArrayList()
176     }
177     val width = size.width.toDouble()
178     val height = size.height.toDouble()
179     var minV = min //lowest value in buffer
180     var maxV = max //highest value in buffer
181     if (minV == maxV) {
182         minV = minV / 2
183         maxV = maxV + minV / 2
184     }
185     val dv = maxV - minV
186     step = width / buffer_.capacity()
187     val coeff = (height - 2 * shiftH) / dv
188     //passing the wrong buffer to dataSeriesOverlay (not updated)
189     val dataTemp =
190         if (_mode == GraphMode.overlay) utils_.dataSeriesOverlay(buffer_) else utils_.dataSeriesNormal(
191             storeWrapper: this
192         )
193     data = ArrayList(Collections.nCopies(dataTemp.size, 0.0))
194     //return data
195     if (data.isEmpty() || dataTemp.isEmpty()) {
196         return data
197     }
198     for (i in dataTemp.indices) {
199         data[i] = (maxV - dataTemp[i]!!) * coeff + shiftH
200     }
201     return data
202 }
203

```

Figure 10: Mapping ECG-Data code

Preparing the Drawpath

To achieve a smooth and natural representation of the ECG waveform, it is essential to create two separate paths: one for the data points before the current write position in the buffer and one for the data points after. This ensures a seamless transition between old and new data, preventing visual gaps and making the waveform appear continuous, just like a real ECG signal.

The function `preparePathBefore(data: List<Double>)` constructs the path for the portion of the data preceding the current write index of the buffer. It determines the correct starting index and iterates through the data points up to the write position, generating a Path object that maps the signal to drawable coordinates.

Conversely, the function `preparePathAfter(data: List<Double>)` handles the data following the write index. It ensures that the path continues from the last drawn point, maintaining a smooth flow of the signal. By splitting the waveform into these two sections, the transition between old and new data is visually imperceptible, preserving the real-time appearance of the ECG signal.

This approach is crucial for maintaining a continuous and realistic ECG display, as it eliminates abrupt jumps or gaps in the graph, ensuring that the visualization remains fluid and accurately represents the ongoing signal dynamics.

```

215 fun preparePathBefore(data: List<Double>): Path {
216     var idx_ = buffer_.writeIndex() - 1
217     if (idx_ >= data.size) {
218         idx_ = data.size - 1
219     }
220 }
221 val idx = if (idx_ < 0) 0 else idx_
222 val path = Path()
223 path.moveTo(x: 0f, data[0].toFloat())
224 for (i in 1 ≤ until < idx - 1) {
225     path.lineTo((i * step).toFloat(), data[i].toFloat())
226 }
227 return path
228 }
229
230 fun preparePathAfter(data: List<Double>): Path {
231     var idx_ = buffer_.writeIndex()
232     if (idx_ >= data.size) {
233         idx_ = data.size - 1
234     }
235     val idx = if (idx_ < buffer_.capacity() - 1) idx_ else buffer_.capacity() - 1
236     val path = Path()
237     path.moveTo((idx * step).toFloat(), data[idx].toFloat())
238     for (i in idx + 1 ≤ until < data.size) {
239         path.lineTo((i * step).toFloat(), data[i].toFloat())
240     }
241     return path
242 }

```

Figure 11: prepare paths functionality code

4.6 GraphEcgWidget

5 SUMMARY

This thesis presents the design and implementation of a frontend application for real-time and long-term monitoring of electrocardiographic (ECG) data, utilizing Bluetooth Low Energy (BLE) communication. The primary objective was to develop an efficient and responsive system capable of handling continuous ECG data streams while ensuring smooth visualization and analysis. Key challenges included managing high-frequency data acquisition, real-time signal processing, and rendering an intuitive graphical representation. By leveraging Kotlin and Jetpack Compose, an optimized architecture was established, incorporating coroutines for concurrent data processing and an adaptive buffer for efficient data storage and retrieval.

6 RESULT

The developed system successfully processes and visualizes ECG data in real time. Through the implementation of a ring buffer and sliding window approach, the application maintains a continuous and smooth representation of the ECG signal, preserving critical waveform characteristics. The use of the Pan-Tompkins algorithm allows for reliable QRS peak detection, enhancing the accuracy of heartbeat identification. Additionally, BLE communication was optimized to minimize latency and packet loss, ensuring stable data transmission. The performance evaluation indicates that the system meets the requirements for real-time ECG monitoring, achieving both responsiveness and scalability.

7 OUTLOOK

Future enhancements could focus on integrating machine learning techniques for advanced arrhythmia detection, further improving the diagnostic capabilities of the application. Additionally, expanding compatibility with multiple BLE-enabled ECG devices could increase the system's versatility. Implementing cloud synchronization for remote monitoring and historical data analysis would further extend the practical applications of this work. Lastly, optimizing power consumption for mobile usage remains a crucial aspect for improving long-term monitoring solutions.

8 APPENDIX

8.0.1 Versions

This section presents fundamental versions of the required development resources used for this work.

The tests for this application during the development process, were done on a phone using the **Android version 13** which implies the usage of **API level 33**. Utilizing lower Android version than 12 (API Level 31,32) is not possible due to new runtime permissions added in Android 13, making it impossible to start the app⁵. Higher Android versions were not reviewed. More versions of the used resources, are presented in table 3.

Table 3: Versions of used resources

Ressource	Version used
Android version	13
API Level	33
Kotlin compiler extension version	5.15.1
minSDK	30

8.1 Beispielcode zur Erzeugung eines Quelltext-Generators

```

1      @SuppressWarnings("MissingPermission")
2      fun connect(context: Context, device: ServerDevice) {
3          this.viewModelScope.launch {
4              val connection = ClientBleGatt.connect(context, device,
5                  this@BLEViewModel.viewModelScope )
6              this@BLEViewModel.connection.value = connection
7              val services = connection.discoverServices()
8              val bleService = services.findService(UUID.fromString("
9                  e28e00e9-b79a-479a-a6f1-21e8f73236e1"))
10             val signalCharacteristic = bleService!!.findCharacteristic(
11                 UUID.fromString("23b85b26-e7cb-421c-9731-1b4ca2d2a849"))
12             val appCommuncationCharacteristic = bleService!!.
                findCharacteristic(UUID.fromString("563a2846-1dc7-11ef
                    -9262-0242ac120002"))
                this@BLEViewModel.appCommuncationCharacteristic =
                    appCommuncationCharacteristic!!

                signalCharacteristic!!.getNotifications().onEach {

```

⁵<https://developer.android.com/develop/connectivity/bluetooth/bt-permissions>

```

13         val values = byteArrayToIntArray(it.value).toList()
14         startTime = System.currentTimeMillis()
15         channel.send(values)
16         receiveData(viewModelScope){dataList ->
17             StoreWrapper.processDataList(dataList)
18             BPM.currList.value += dataList
19             if(BPM.currList.value.size >= 7500){
20                 var qrsPeaks = BPM.panTompkins(BPM.currList.
21                     value).size
22                 println("PEAK COUNTER: ${qrsPeaks}")
23                 println("liste: ${BPM.currList.value}")
24                 BPM.currList.value.forEach {
25                     println(it)
26                 }
27                 println("ENDOF PRINTING")
28                 BPM.currList.value = BPM.currList.value.subList
29                     (750, 7500-1)
30                 BPM.countPeak = qrsPeaks
31                 qrsPeaks = 0
32             }
33         }
34         paketCounter++
35     }.launchIn(viewModelScope)
36     appCommunicationCharacteristic!!.getNotifications().onEach {
37         ResponseChecker.check(it.value)
38         PacketReassembler.reassemble(it.value)
39         println(it)
40     }.launchIn(viewModelScope)
41     connection.requestMtu(512)
42     fetchCurrentMode()
43     listEKGS()
44     750
45 }
46 }

```

Code 2: Einfaches Beispiel eines Sourcecode-Generators, abgeleitet von ?? und eigenen Ergänzungen

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